

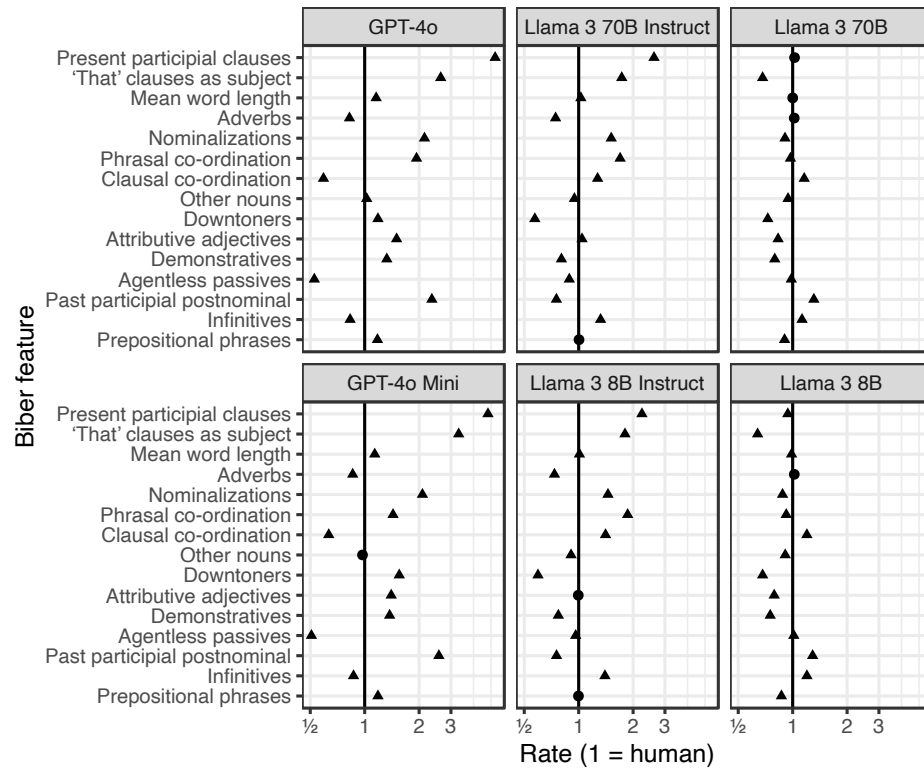


Fig. 3. Rate of Biber feature use by different LLMs, relative to the human usage of each feature, for the top 15 most important features in the HAP-E corpus. Note the log scale. GPT-4o and GPT-4o Mini show the largest variation from human texts, while the base variants of Llama 3 most closely resemble human grammar and style. Larger models (top row) generally show the same stylistic differences as their smaller counterparts (bottom row), despite performing better on other benchmark tasks. Triangles indicate statistically significant differences from human usage. Figure S1 of the SI Appendix gives the distributions of paired differences.

Table 1. Most overrepresented words in LLM-generated texts, relative to human usage rates

GPT-4o		GPT-4o Mini		Llama 3 70B Instruct		Llama 3 8B Instruct		Llama 3 70B		Llama 3 8B	
Word	Rate	Word	Rate	Word	Rate	Word	Rate	Word	Rate	Word	Rate
camaraderie	162	camaraderie	171	unease	63	unease	101	bananas	31	deborah	52
tapestry	155	tapestry	147	palpable	47	continuation	52	paperback	30	rambo	22
intricate	119	palpable	145	continuation	29	palpable	48	bam	26	matty	20
underscore	107	grapple	131	shoutout	28	reminder	33	verona	25	goodnight	18
unspoken	102	intricate	129	intricate	27	pang	29	filth	19	ml	15
amidst	100	fleeting	124	pang	25	rut	29	rekall	17	merlin	13
palpable	95	ignite	122	camaraderie	24	waft	28	denis	14	worcester	11
solace	95	vibrant	92	policy maker	24	prioritize	27	darry	12	fay	10
fleeting	84	amidst	90	prioritize	24	grapple	24	ebook	12	missy	10
unravel	83	cacophony	89	reminder	24	camaraderie	23	janice	12	elisa	10

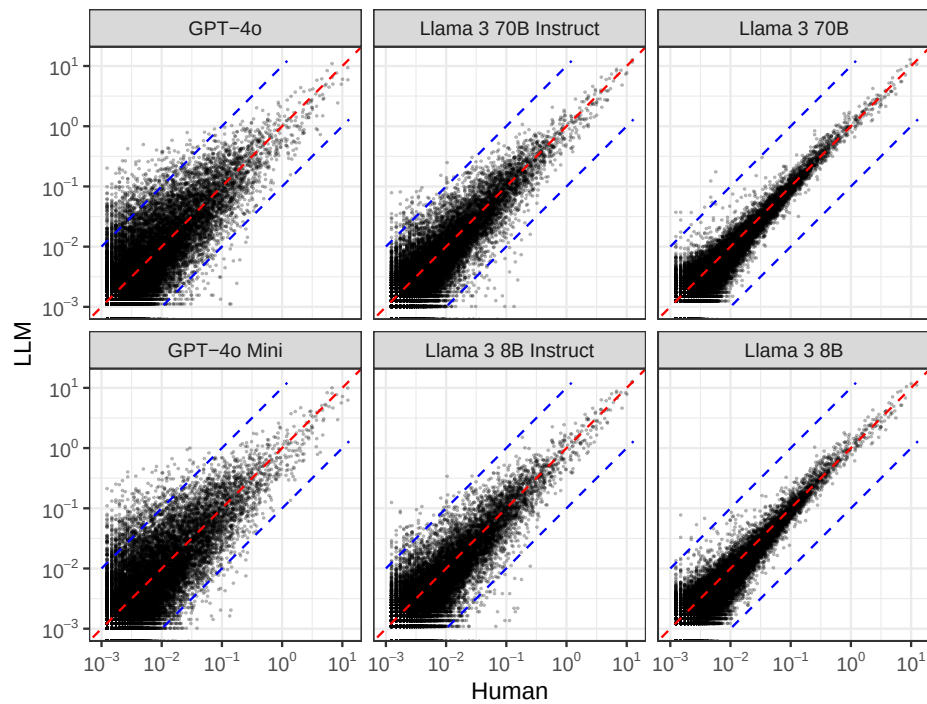


Fig. 4. Rates of word use by different LLMs (per 1,000 words) compared to the human use of each word in chunk 2, in the HAP-E corpus (log scale). Includes all words used more than once per million words in chunk 2. Words are lemmatized to group together inflected forms. Words on the diagonal are used equally often in human and LLM texts. Dashed blue lines indicate the range between $10\times$ more and $10\times$ less than human use. Note that the instruction-tuned models show more variation from the diagonal, indicating more deviation in vocabulary use relative to humans.

Discussion

This study identifies salient differences both between human and LLM-generated texts and among various models. The features that distinguish between humans and different LLMs include present participial clauses, ‘that’ clauses as sentence subject, passive voice, and nominalizations, to name a few. These findings corroborate other research that points out the ways these features produce informationally dense prose (28). Other research links these features to increased lexical diversity in generated text, as well as human judgments of linguistic mastery (13). Lastly, prior work found that ChatGPT-4 text evidences more nominalizations, and fewer human subjects and epistemic stance markers (29), findings we see reproduced in our list of distinguishing features.

A second major finding of this research is the apparently central role of instruction tuning in creating these discrepancies between human and model general texts. While we do not have access to untuned versions of GPT, comparisons between Llama’s base and tuned models emphasize the degree to which instruction tuning pushes models to produce text that reads *unlike* a human. This suggests that differences in style are not simply due to the selection of texts for training the base models, but due to the instruction-tuning process. Similarly, differences between GPT-4o and the instruction-tuned Llama variants may be due to differences in instruction tuning, either through different human preferences in rating responses or differences in the tasks (such as summarization) used to tune the models. (As the instruction tuning processes are not publicly documented, it is not possible to determine the cause more precisely.) While instruction tuning has previously been shown to introduce cognitive biases (30), to our knowledge, these changes in writing style are not discussed elsewhere in similar research.

A third major finding of this work is significant success of Biber’s tagset in modeling and classifying text. This success suggests that varied linguistic perspectives—which, perhaps, are not prioritized during the development and in-house assessments of LLMs—can reveal otherwise tacit information that distinguishes a text as machine-generated. With the linguistic perspective offered by pseudobibeR, we built a model that recognizes machine-generated text with relative ease. Our study reveals the clear value of linguistics expertise and functional conceptions of language in both LLM use and development.

That said, our intention is not to propose another way to construct LLM detectors or to police the writing of students and learners. Instead, we maintain that this type of comparative analysis is useful for identifying differences between human- and machine-generated text, zeroing in on specific teachable moments in the revision of machine-generated text.

As LLMs are increasingly put to work completing diverse writing tasks, these results suggest a notable misalignment between generated texts and the contexts in which we put them to use. This is another way of saying that LLMs do not vary their linguistic output in response to contextual factors in ways similar to humans. This misalignment affects experts and learners differently. For those proficient in a genre—think, a therapist collating notes on a patient or, say, a college graduate writing thank-you notes to friends and family—this misalignment is likely flagged and the output is

appropriately revised. When the writer is proficient in the genre, previous experience guides a current sense of what a particular document should look like in order to be successful. For experts, then, LLMs appear a worthwhile productivity tool, suitable so long as they lend their expertise to further shaping the output.

For learners, though, LLMs appear more problematic. Of course, using LLMs to learn more about a concept, or help generate ideas, is one thing. Using output in a text is a different matter, one that may affect a student’s learning trajectories. In this case, students offload the important cognitive labor of shaping a text for a particular audience and purpose to the LLM. Never mind that LLMs do not appear to write like humans—when students offload writing work, they offload opportunities to learn how to write IMRaD articles, client reports, executive summaries, investment pitches, etc. When LLMs are used in the classroom as writing tools, instructors of all levels and disciplines need to help students see both the shortcomings of the generated text and avenues for improvement. LLMs are not bad, either technically or morally—it is only that instructors must help inculcate in students the critical perspective of the expert to know what’s working and what isn’t.

In other contexts, however, overreliance on LLMs could produce output that might be awkward and inauthentic (e.g., in a creative genre), confusing (e.g., in instructional material), or unpersuasive (e.g., in argumentative texts). The current work thus suggests the importance of LLM practice—both in and out of the classroom—informed by human expertise via a continual dialogue of creation and revision, where LLM users are more aware and mindful of the effective uses as well as the limitations of various LLMs.

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